
Do Tabular Foundation Models Scale with Dimensionality?

A Synthetic-Data Study from 10 to 10,000 Features

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Abstract

TODO: fill in after results are final. One paragraph: motivation, method (synthetic low-dimensional nonlinear signal embedded in up to 10,000 features, feature-bagging ensemble for models beyond their native feature cap), and the headline empirical finding (how TabPFN v2 / TabICL test R^2 degrades relative to tree baselines as D grows, under fixed- N and N -scaling-with- D regimes).

1. Introduction

Tabular foundation models – transformer architectures pre-trained on large corpora of synthetic or real tabular datasets and applied zero-shot via in-context learning – have shown striking accuracy on small- to medium-sized tabular regression and classification benchmarks (Müller et al., 2022; Hollmann et al., 2025). Unlike gradient-boosted trees or linear models, these models are not refit per dataset; instead, the training set is passed as context alongside the query points, and a single forward pass produces predictions.

This convenience comes with an architectural constraint that has received little systematic study: every released tabular foundation model has a hard cap on the number of features and rows it was pretrained and validated on. TabPFN v2 (Hollmann et al., 2025) was trained and evaluated up to 500 features and 10,000 rows; TabICL (Qu et al., 2025) extends this considerably but still caps out in the low thousands of features. Real tabular data routinely exceeds these limits – genomics, click-stream, and sensor-fusion datasets can have thousands to tens of thousands of columns – so understanding *how* these models degrade (or don’t) past their trained regime, and whether standard workarounds close the gap, is a practical question for anyone considering a tabular foundation model outside a benchmark setting.

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We study this question with a controlled synthetic data generating process: a fixed, low-dimensional (5-feature) nonlinear function determines the regression target, and the remaining $D - 5$ ambient features are either noisy nonlinear functions of the true features or pure noise, with D swept on a log-spaced grid from 10 to 10,000. Because the true signal’s dimensionality is fixed while the ambient dimensionality grows, this isolates the effect of dimensionality *per se* from the effect of adding more true signal, which is the confound present in most real high-dimensional tabular benchmarks. We compare TabPFN v2 and TabICL, each wrapped in a feature-bagging ensemble (Feuer et al., 2023) once D exceeds their native cap, against Ridge regression, Random Forest (Breiman, 2001), and XGBoost (Chen & Guestrin, 2016), under both a fixed training-set-size regime and a regime where training set size scales with D .

Our contributions are:

- A synthetic benchmark generator with a controllable, low-dimensional nonlinear true signal embedded in up to 10,000 ambient features, released with this paper.
- An empirical characterization of how held-out test RMSE/ R^2 for TabPFN v2 and TabICL scale with D , relative to tree and linear baselines, under both fixed- N and N -scaling-with- D regimes.
- A direct empirical test of the feature-bagging ensemble workaround (Feuer et al., 2023) at the specific scale (10,000 features) it was proposed to address.

2. Related Work

Tabular foundation models. Müller et al. (2022) introduced Prior-Data Fitted Networks (PFNs), transformers trained on synthetic data to approximate Bayesian inference in a single forward pass, which Hollmann et al. (2025) scaled into TabPFN v2, a model pretrained once and applied zero-shot to new tabular datasets via in-context learning, achieving Nature-cover accuracy on datasets up to 10,000 rows and 500 features. Qu et al. (2025) proposed TabICL, which restructures the in-context learning architecture (a column-then-row transformer with a fixed-size latent summary) specifically to scale to far larger row counts

and a higher feature cap. Ma et al. (2024) (TabDPT) and Zeng et al. (2025) (TabFlex) pursue related scaling directions – retrieval-augmented real-data pretraining, and linear-attention mechanisms, respectively – motivated by the same observation we study directly: that naively applying an in-context tabular learner outside its trained regime degrades performance.

Scaling TabPFN past its trained regime. Feuer et al. (2023) is the paper most directly relevant to our method: they show TabPFN’s accuracy falls as feature count grows past its trained regime, and propose closing the gap with sketching, feature selection, and feature-bagging ensembles – fitting independent copies of TabPFN on random feature subsets and averaging. We adopt their feature-bagging approach as the mechanism for applying TabPFN v2 and TabICL beyond their native caps, and, unlike their work (which evaluates on existing high-dimensional tabular benchmarks, where the number of truly informative features is unknown and grows with D), we hold the number of truly informative features fixed at 5 while sweeping D , isolating the effect of ambient dimensionality from the effect of additional signal.

High-dimensional tabular learning more broadly. Feature selection, random projections, and ensembling over feature subsets are classical tools for high-dimensional tabular learning with tree and linear models; our contribution is not a new such tool, but an empirical measurement of how well the existing tool (feature-bagging) closes the dimensionality gap for the specific architectures that motivated it.

3. Method

3.1. Synthetic data generating process

Each dataset is parameterized by an ambient feature count $D \geq 5$ and a sample count N . We draw $D_{\text{true}} = 5$ latent features $z = (z_1, \dots, z_5) \sim \mathcal{N}(0, I_5)$ per row and define the regression target through a fixed nonlinear function:

$$f(z) = 3 \sin(z_1 z_2) + 2z_3^2 - 1.5 z_4 z_5 + z_1 z_4^2 - 0.5 z_2^3 + 2 \tanh(z_1 + z_3 - z_4), \quad (1)$$

$$y = f(z) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, (\eta \cdot \sigma_f)^2), \quad (2)$$

where σ_f is the empirical standard deviation of $f(z)$ over the batch and η (the noise-to-signal ratio) is fixed across all D so the achievable ceiling on test R^2 does not itself vary with dimensionality.

The remaining $D - 5$ ambient features are split into two kinds. A fraction p_{noise} are pure noise, $x_j \sim \mathcal{N}(0, 1)$, independent of y . The rest are noisy derived features: for each, we sample $k \in \{1, 2, 3\}$ of the true latent features uniformly without replacement, a random weight vector,

and one of four transforms (identity, square, sine, tanh) applied to the weighted sum, then add independent Gaussian noise at scale ρ . These derived features are correlated with y (through z) but not perfectly recoverable, matching the informal specification that ambient features be “correlated or noisy functions of the true features...but not 100% derivable.” All D columns (true, derived, and pure-noise) are then randomly permuted so column position carries no information. Train, validation, and test splits are drawn as one i.i.d. block per (D, N, seed) and sliced, since rows are independent draws.

3.2. Dimensionality and sample-size grid

We sweep $D \in \{10, 20, 50, 100, 200, 500, 1000, 2000, 5000, 10000\}$. Two regimes control the training set size N_{train} : **fixed-N**, where N_{train} is held at a constant TODO across all D ; and **scaling-N**, where $N_{\text{train}} = \text{clip}(4D, \text{TODO}, \text{TODO})$. Validation and test set sizes (TODO, TODO) are held fixed across all conditions so evaluation is comparable. Each cell is repeated over TODO random seeds.

3.3. Models

We evaluate Ridge regression (Hoerl & Kennard, 1970), Random Forest (Breiman, 2001), XGBoost (Chen & Guestrin, 2016), TabPFN v2 (Hollmann et al., 2025), and TabICL (Qu et al., 2025). For the two foundation models, whenever D exceeds the model’s native feature cap ($c = 500$ for TabPFN v2, $c = 2000$ for TabICL), we apply a feature-bagging ensemble (Feuer et al., 2023): partition the D columns into $\lceil D/c \rceil$ disjoint chunks of size $\leq c$ via a random permutation, fit an independent copy of the model on each chunk, and average the chunks’ predictions. When $D \leq c$ this reduces to a single chunk containing every feature, i.e. the model runs exactly as it natively would.

4. Experimental Setup

TODO: hardware/software (Apple Silicon CPU, no CUDA), library versions, and any model-specific settings (e.g. `ignore_pretraining_limits=True` for TabPFN so a single chunk that exceeds its internal threshold does not raise rather than degrade).

5. Results

TODO: figures (figures/r2_vs_dimensionality.pdf, figures/rmse_vs_dimensionality.pdf) and summary tables, once the sweep has run to completion.

6. Discussion and Limitations

TODO. Points to cover regardless of the empirical outcome:

- The feature-bagging ensemble is an approximation we impose on TabPFN/TabICL, not a capability native to either model; results past each model’s native capabilities characterize *our wrapper*, not the released model in isolation.
- All experiments ran on CPU (Apple Silicon, no CUDA); TabPFN v2/TabICL are primarily validated and optimized for GPU inference, so absolute wall-clock numbers here are not representative of production latency.
- TabPFN v2 and later PriorLabs checkpoints are released under a non-commercial research license; TabICL is BSD-3-Clause.
- A single fixed nonlinear true function and one specific derived-feature construction were used; results characterize this generative family, not dimensionality scaling in general.

7. Conclusion

TODO.

Acknowledgements

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References

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